

A binding site for phosphoinositide modulation of voltage-gated sodium channels described by multiscale simulations

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Abstract

Voltage gated sodium channels (Na_v) are membrane proteins which open to facilitate the inward flux of sodium ions into excitable cells. In response to stimuli, Na_v channels undergo a transition from the resting, closed state to an open state which allows ion influx, before rapidly inactivating. Dysregulation of this functional cycle due to mutations leads to diseases including epilepsy, pain conditions and cardiac disorders, making Na_v channels a significant pharmacological target. Phosphoinositides are important lipid cofactors for ion channel function. The phosphoinositide $\text{PI}(4,5)\text{P}_2$ decreases $\text{Na}_v1.4$ activity by increasing the difficulty of channel opening, accelerating fast activation and slowing recovery from fast inactivation. Using multiscale molecular dynamics simulations, we show that $\text{PI}(4,5)\text{P}_2$ binds stably to inactivated Na_v at a conserved site within the DIV S4-S5 linker, which couples the voltage sensing domain (VSD) to the pore. As the Na_v C-terminal domain is proposed to also bind here during recovery from inactivation, we hypothesise that $\text{PI}(4,5)\text{P}_2$ prolongs inactivation by competing to bind to this site. In atomistic simulations, $\text{PI}(4,5)\text{P}_2$ reduces the mobility of both the DIV S4-S5 linker and the DIII-IV linker, responsible for fast inactivation, slowing the conformational changes required for the channel to recover to the resting state. We further show that in a resting state Na_v model, phosphoinositides bind to VSD gating charges, which may anchor them and impede VSD activation. Our results provide a mechanism by which phosphoinositides alter the voltage dependence of activation and the rate of recovery from inactivation, an important step for the development of novel therapies to treat Na_v -related diseases.

eLife assessment

This study employs multiscale simulations to show that PIP2 lipids bind to the DIV S4-S5 linker within the inactivated state of a voltage-gated sodium channel, affecting the coupling of voltage sensors to the ion-conducting pore. The authors report the **valuable** finding that PIP2 prolongs inactivation by binding to the same site that binds the C-terminal tail during recovery from inactivation, suggesting binding to gating charges in the resting state that may impede activation. The coarse-grained and atomistic molecular dynamics simulations are **solid**, supporting the claims, although they do lack validation through experiment or simulation, such as studies of linker mutants to confirm PIP2 binding sites, a resting state Nav1.4 model to confirm voltage sensor binding, as well as tests using flexible linkers to probe interactions with PIP2 without backbone restraints.

Introduction

Voltage-gated sodium (Na_v) channels are critical to the regulation of brain activity, cardiac rhythm, and muscle contraction. Expressed in the membranes of excitable cells, Na_v channels respond to membrane depolarization to open a pore that facilitates the selective flow of sodium current into the cell, initiating the action potential. In mammals, the Na_v channel family consists of nine subtypes (Na_v 1.1-1.9), distributed throughout the central and peripheral nervous system, as well as in cardiac and skeletal muscle (Yu and Catterall, 2003 [\[1\]](#)). The Na_v 1.4 subtype is predominantly expressed in skeletal myofibers, where it initiates muscle contraction. Genetic mutations in this subtype are associated with various motor dysfunctions, such as both hyperkalaemic and hypokalaemic periodic paralyses (Jurkat-Rott et al., 2000 [\[2\]](#); Mantegazza et al., 2021 [\[3\]](#); Venance et al., 2006 [\[4\]](#)). Na_v 1.7 is found in peripheral sensory neurons and is responsible for nociception. Several pain disorders, such as inherited erythromelalgia and small fibre neuropathy arise from gain-of-function Na_v 1.7 mutations (Dib-Hajj et al., 2013 [\[5\]](#)). Both Na_v subtypes have been investigated as promising pharmacological targets for the treatment of myopathy and pain conditions (Dib-Hajj et al., 2013 [\[5\]](#); Ghovanloo et al., 2021 [\[6\]](#)).

Structurally, Na_v channels consist of four homologous domains (DI-DIV) arranged in a domain-swapped configuration (Fig. 1A-B [\[7\]](#)). Each domain comprises of six transmembrane helices (S1-S6). The central pore domain is formed by S5, S6 helices and selectivity filter (SF), while the four peripheral voltage-sensing domains (VSDs) are formed by S1-S4 (de Lera Ruiz and Kraus, 2015 [\[8\]](#); Pan et al., 2018 [\[9\]](#)). The pore domain also features lateral fenestrations that provide a pathway for the access of small molecules to the pore via the membrane (Gamal El-Din et al., 2018 [\[10\]](#)) and have been shown in computational studies to be accessible to lipid tails (Boiteux et al., 2014 [\[11\]](#); Martin and Corry, 2014 [\[12\]](#); Raju et al., 2013 [\[13\]](#); Tao and Corry, 2022 [\[14\]](#)). Additionally, the C-terminal domain (CTD) extends from the DIV S6 helix into the cytoplasm, where it is thought to associate with the DIII-IV and DIV S4-S5 linkers in the resting state (Clairfeuille et al., 2019 [\[15\]](#)).

Na_v channels adopt distinct functional states during the cycle of membrane depolarization and repolarization (Fig. 1C [\[16\]](#)). In the resting state, the pore is closed and VSD S4 helices are in the down, deactivated state. Membrane depolarization triggers the asynchronous transition of VSD I-III S4 helices to the up, activated conformation, causing Na_v channels to open (Capes et al., 2013 [\[17\]](#); Cha et al., 1999 [\[18\]](#); Goldschen-Ohm et al., 2013 [\[19\]](#)). During prolonged depolarization, VSD-IV moves upwards, causing the channels to adopt a fast inactivated state in which a three-residue (IFM) hydrophobic motif located on the intracellular linker between DIII and DIV (DIII-IV linker) allosterically closes the pore (Yan et al., 2017 [\[20\]](#)). The CTD is proposed to bind at the S4-S5 linker of

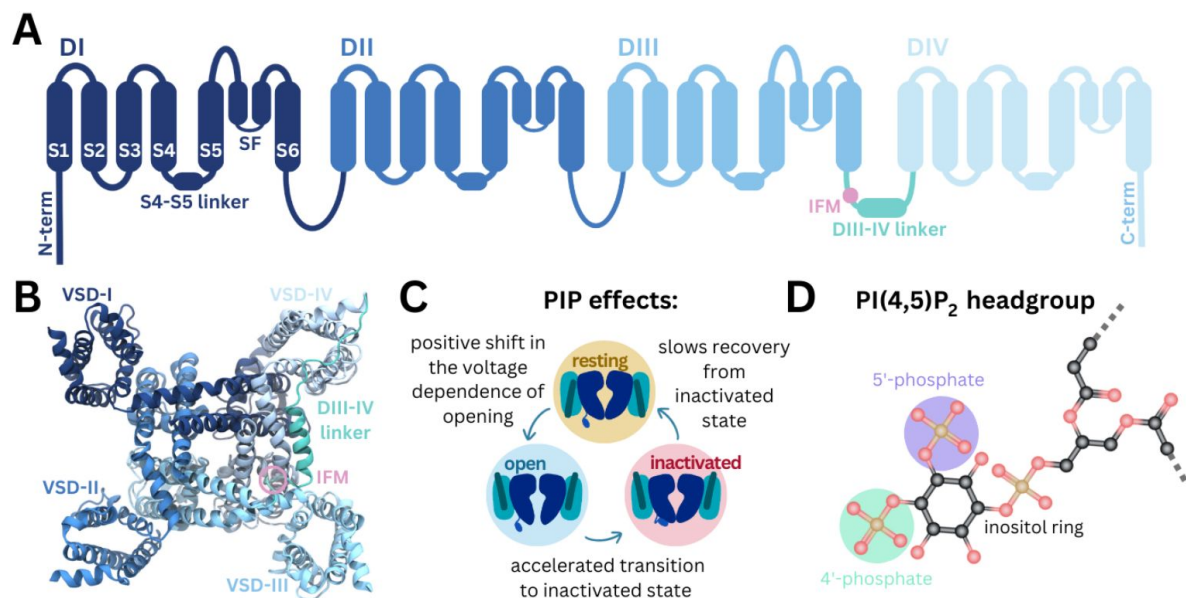


Fig. 1.

Structure of Na_v and modulation by phosphoinositides

(A) Na_v channel topology featuring transmembrane helices (S1-S6), the selectivity filter (SF) and the DIII-IV linker (containing the IFM motif) located between DIII and DIV. **(B)** Na_v1.4 structure (6agf) showing the four domain-swapped voltage sensor domains (VSDs I-IV), pore and DIII-IV linker on the intracellular side. **(C)** Summary of PI(4,5)P₂ effects on transitions between Na_v channel functional states (Gada *et al.*, 2023 [eLife.91218.1](https://doi.org/10.7554/eLife.91218.1)) **(D)** Structure of the PI(4,5)P₂ headgroup with the 4'- and 5'-phosphates indicated

VSD-IV and sequester the DIII-IV linker during the resting state. During fast inactivation, the CTD dissociates from VSD-IV and releases the DIII-IV linker to allow IFM binding (Clairfeuille *et al.*, 2019). Upon repolarization, the VSDs deactivate, the IFM motif disassociates, the CTD rebinds to VSD-IV and the pore returns to the resting state.

Phosphoinositides (PIPs) are important cellular signaling molecules found on the cytoplasmic leaflet of the mammalian cell membrane. They can exist in seven forms, with phosphorylation possible at one (PIP1), two (PIP2) or all three (PIP3) positions on the inositol ring, at the 3', 4' and/or 5' carbons. PIPs, particularly PI(4,5)P₂, featuring phosphates at the 4' and 5' carbon positions (Fig. 1D), are known to bind and modulate the activity of numerous ion channels families (Hille *et al.*, 2015). These include voltage-gated ion channels, some of which have been resolved with PI(4,5)P₂ bound (Gao *et al.*, 2021; Sun and MacKinnon, 2020). PIP is known to interact with the VSDs of different potassium channels, to stabilize the positive gating charges and support the voltage-sensing mechanism (Schmidt *et al.*, 2006). By also binding at the VSD-pore interface in channels such as K_v7.1, PIP is proposed to facilitate coupling of VSD movement to pore opening (Kasimova *et al.*, 2015; Ma *et al.*, 2022; Sun and MacKinnon, 2020). PI(4,5)P₂ also forms specific interactions with VSD-II of Cav2.2 in the down-state, making channel activation more difficult (Gao *et al.*, 2021). Although PI(4,5)P₂ is known to bind to numerous voltage gated ion channels, its effects on channel gating and function are complex and yet to be fully elucidated.

Recent experiments show that Na_v1.4 channel kinetics and voltage dependence are modulated by PI(4,5)P₂ (Gada *et al.*, 2023). PI(4,5)P₂ inhibits Na_v1.4 by causing a depolarizing shift in voltage dependence of activation, accelerating transition to the inactivated state and slowing recovery from inactivation, resulting in reduced peak current and suppression of late current (summarized in Fig. 1C). While this is likely to occur via a direct interaction with Na_v1.4, the structural basis of PI(4,5)P₂ modulation remains to be understood.

Here, we used a combination of coarse-grained and atomistic MD simulations to identify a putative PIP binding site to inactivated Na_v1.4 in VSD-IV and the DIII-IV linker. We analyze the atomistic level interactions between the positively charged residues at this site with PI(4,5)P₂ and PI(4)P, comparing this with structurally resolved PIP binding sites in the related ion channels Ca_v2.2 and K_v7.1. Consistent with the sequence conservation at the identified site, we find that PIPs also bind to Nav1.7 in coarse-grained simulations, with notable differences dependent on VSD conformation states. This work provides insight into how PIPs can negatively regulate Na_v channels, a first step for the potential development of PIP-analogue sodium channel inhibitors.

Results

To investigate how diverse lipid species interact with Na_v1.4, we carried out coarse-grained simulations of Na_v1.4 (PDB ID: 6agf, inactivated state) embedded in a complex mammalian membrane for 16 μs in triplicate (Fig. 2A). Glycosphingolipid (GM), phosphoinositide (PIP) and diacylglycerol (DAG) were highly enriched around Nav1.4 (Fig. 2B). Additionally, we observed modest enrichment of lysophosphatidylcholine (LPC), phosphatidylinositol (PI), phosphatidylserine (PS) and phosphatidylethanolamine (PE) and slight depletion of ceramide (CER), sphingomyelin (SM), phosphatidylcholine (PC) and cholesterol (CL). To investigate specific interactions, we generated z-density maps and calculated the per-residue occupancy of the 12 different lipid types (Fig. 2C, Fig. S1-2). Binding residues of interest were identified by constructing occupancy distributions by residue for each lipid type and identifying outlying values with high occupancies (Fig. S3). DG lipids form significant interactions within the lateral fenestrations of Na_v1.4 (Fig. S2). LPC and PI also frequently interact with different VSD residues (Fig. S1).

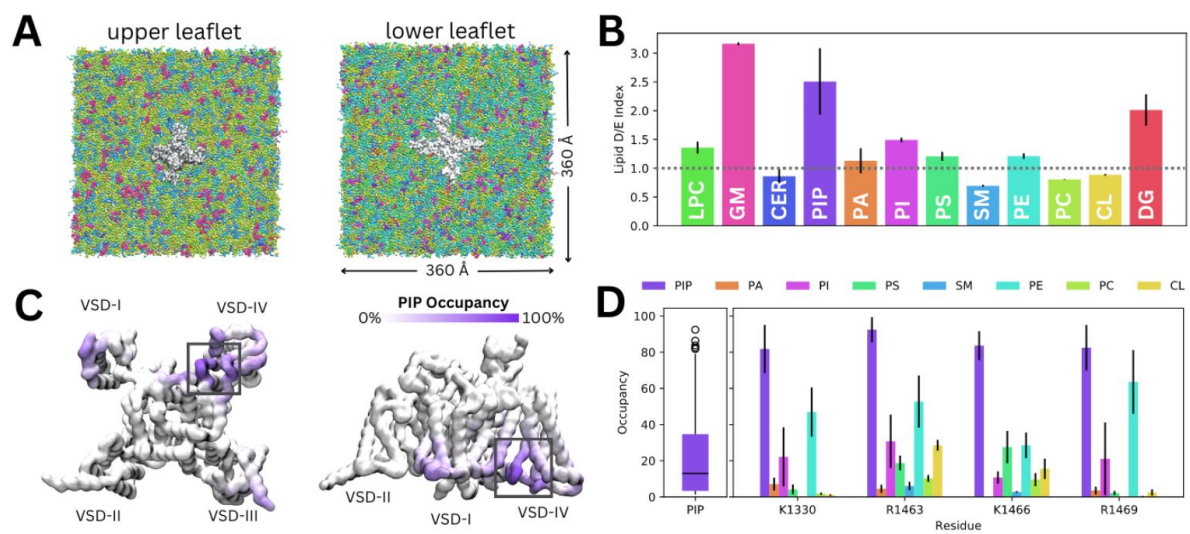


Fig. 2.

Lipid fingerprint and binding of all PIP types to Na_v1.4

(A) Na_v1.4 embedded in a 360 Å x 360 Å model mammalian membrane containing 63 lipid species. (B) Lipid depletion enrichment index of lipids around Na_v1.4 grouped into 12 headgroup types (C) Na_v1.4, shown from the intracellular (left) and membrane (right) sides, colored by PIP occupancy (darker purple = greater PIP occupancy) (D) Distribution of PIP binding occupancies (left) and occupancy of lipid species at 4 residues with the highest PIP occupancy.

Given our interest in understanding the modulation of Na_v1.4 by PIPs, we focus on their interactions for the remainder of this manuscript. Despite the very low concentration of PIPs (0.5% each of PIP1, PIP2 and PIP3) in the mammalian membrane, they are highly enriched around Na_v1.4, particularly near the DIII-IV linker and the VSD-IV (**Fig. 2C**). Contact analysis revealed a putative PIP binding site involving the K1330 residue of the DIII-IV linker and residues R1463, K1466 and R1469 in the DIV S4-S5 linker, which connects the pore and VSD-IV (**Fig. 2D**). Across all three replicates, only residues within this site were occupied by PIPs for more than 80% (Quartile 3 + 1.5 x Interquartile range) of simulation time on average (**Fig. 2D**). We note that these residues have higher occupancies for PIP compared to other lipids, including other negatively charged phospholipids (PA, PS and PI) (**Fig. 2D**).

Simulations in a complex mammalian membrane showed that PIP species bind specifically and selectively to Na_v1.4 in the presence of other negatively charged lipids and at low, physiological concentrations. However, the large membrane required, and long PIP binding durations prevented sampling of large numbers of binding and unbinding events. To address this, we carried out additional simulations where Na_v1.4 was placed in a smaller POPC membrane with a 5% concentration of each PIP species in the lower leaflet (**Fig. 3A**).

Across ten replicates of these enriched PIP simulations, each carried out for 80 μs, we observed all three PIP species binding at the identified site on VSD-IV and DIII-IV linker (**Fig. 3B**). PIPs can approach and bind to this site from either side of the VSD, however, PIP1 only forms stable interactions when it approaches and binds from the VSD-I side while PIP2 and PIP3 usually bind from the VSD-III side (**Fig. 3B**). These enriched PIP simulations also revealed additional positively charged residues in the DIII-IV linker (K1329, K1333 and K1352) and DIV S4 (R1460) which support binding.

There were 156 PIP binding/unbinding events with duration greater than 2 μs occurring the identified site (**Fig. 3C**). Of these, 43 were with PIP1, 44 with PIP2 and 33 with PIP3. The number of short-term (2-10 μs) interactions decreased with headgroup charge. That is, PIP1 formed the greatest number of short-term interactions while PIP3 had the fewest. Of the 31 binding events with duration greater than 10 μs, 7 were with PIP1, 15 with PIP2 and 9 with PIP3 (**Fig. S3-4**).

When we analyzed interactions occurring during 2-80+ μs binding events, we found that the number of interacting basic residues changes depending on the PIP headgroup charge (**Fig. 3D**). PIP1 (headgroup charge: -3e) binding is most frequently coordinated by two positive charges in VSD-IV and zero or one residue in the DIII-IV linker. PIP2 (headgroup charge: -5e) binding most frequently involves one or two interactions from the DIII-IV linker and three from VSD-IV for a total of five interactions. Interestingly, despite its greater negative charge, PIP3 (headgroup charge: -7e) has interacts similarly to PIP1, and has fewer interactions than PIP2.

The minimum distance between the interacting PIP headgroup and each binding residue across simulation time is shown for the five 40+ μs PIP binding events (**Fig. 3E**). Of these, three are PIP2 binding events which almost span the entire 80 μs of simulation. One 40+ μs binding event each for PIP1 and PIP3 were also observed. The stable PIP1 binding event observed involved interactions with R1460 and R1469 in VSD-IV, as well as fluctuating interactions with the four residues of the DIII-IV linker (K1329, K1330, K1333, K1352). The three long term PIP2 binding events observed were similar to each other, mainly stabilized by R1460, R1463 and R1469. As with PIP1, the number and identity of interacting DIII-IV linker residues varied across the span of each simulation and between replicates, owing to the flexibility of the linker. Like PIP2, PIP3 binding was characterized by stable interactions with R1460, R1463 and R1469 (VSD-IV) as well as K1329 and K1330 (DIII-IV linker).

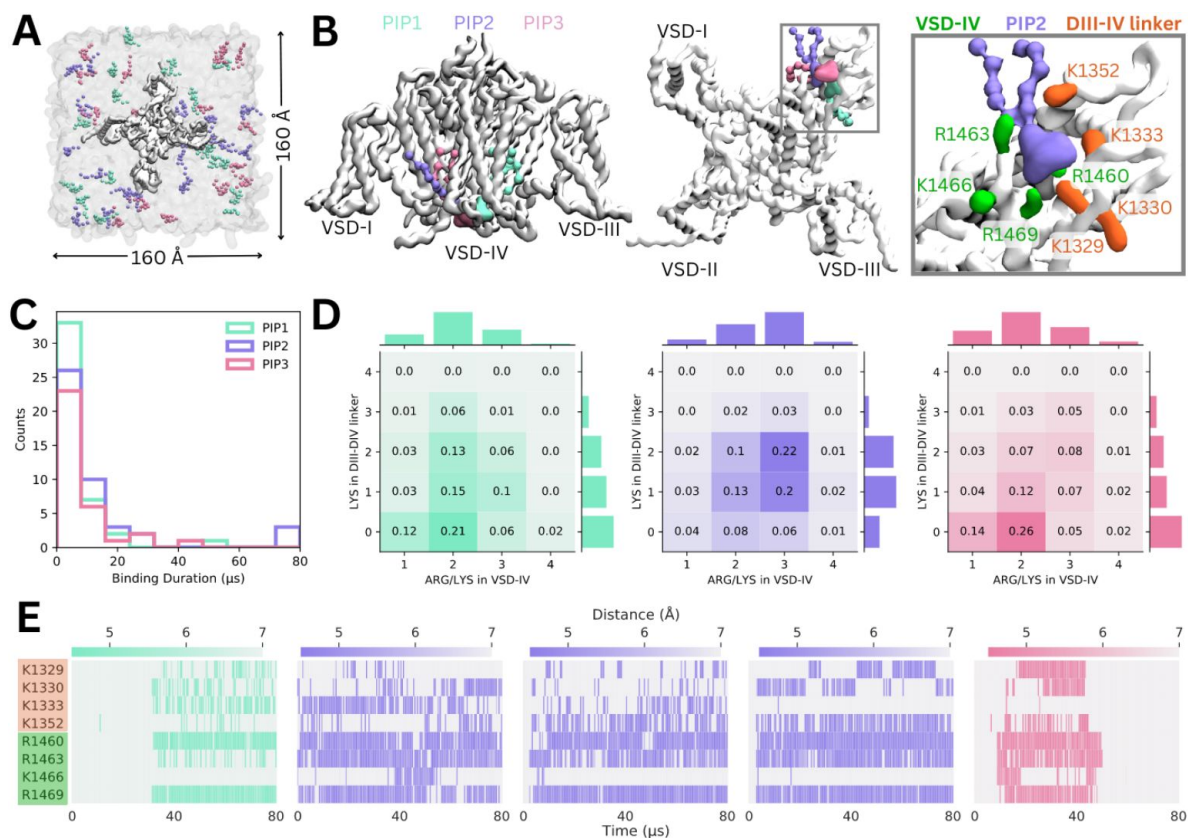


Fig. 3.

Binding of different PIP species in enriched PIP simulations

(A) Enriched PIP simulation system, with Na_v1.4 embedded in a POPC membrane (transparent gray) and 5% each of PIP1 (blue), PIP2 (purple) and PIP3 (pink) added to the lower leaflet. **(B)** Representative snapshots showing PIP binding to VSD-IV and the DIII-IV linker. Na_v1.4 is shown in white and PIPs are colored according to residue name (PIP1 in blue, PIP2 in purple and PIP3 in pink). **(C)** A frequency distribution showing interaction times for each PIP species, defined as the length of a continuous period in which a PIP was within 0.7 nm of two VSD IV binding site residues. **(D)** Frequency plots showing number of positive residues interacting with bound PIP in the DIII-IV linker (vertical) and VSD-IV (horizontal). **(E)** Minimum distance between binding residues on Na_v1.4 and bound PIPs lipid across simulation time for the 5 longest binding events, colored by distance and the type of PIP bound.

Coarse-grained simulations enhance sampling by reducing system size, enabling larger time steps and construction of a smoother energy surface. However, this loss of resolution prevents distinction between phosphate positions on the inositol group and does not permit analysis of protein conformational changes induced by PIP binding. To address this, we backmapped representative snapshots from our PIP enriched simulations, where we observed stable, long-term binding events between Na_v1.4 and PIP1/PIP2. The coarse-grained PIP1 and PIP2 molecules were converted to the atomistic PI(4)P and PI(4,5)P₂ respectively. For each system, 7.5 μs of atomistic simulations were performed (5 replicates, 1.5 μs each).

In these simulations, the PI(4,5)P₂ headgroup was stable at the binding site identified in coarse-grain simulations (RMSD <2.2 Å) (**Fig. 4A** [↗](#), Fig. S6). PI(4,5)P₂ binding is predominantly coordinated by R1469 on the DIV S4-S5 linker, as well as R1466 and R1463. In one replicate, PI(4,5)P₂ associates with R1460 via the phosphate group (PO4) that connects the headgroup to the PIP tails. The PI(4,5)P₂ headgroup also forms electrostatic interactions with K1329 and K1330 in the DIII-IV linker, but forms few contacts with K1333 (**Fig. 4B** [↗](#), Fig. S7). This loop portion of the DIII-IV linker is highly flexible (Fig. S6), thus the lysine sidechains are able to flip between binding the PI(4,5)P₂ headgroup and facing the intracellular space.

The 4'-phosphate formed interactions with residues belonging to the DIII-IV linker (K1329 and K1333) and DIV S4-S5 linker (K1666 and R1469) with similar frequency, in 55-60% of simulation frames (**Fig. 4B** [↗](#)). In contrast, the 5'-phosphate formed contacts with three VSD-IV S4-S5 residues (R1463, K1466 and R1469) in 70-98% of simulation frames and DIII-IV linker residues in 37-48% of frames. Taken together, this data suggests that although the headgroup is flexible when bound, the 5'-phosphate is more important for coordinating VSD-IV S4-S5 residues while the 4'-phosphate associates with both regions of the binding site.

Atomistic simulations of PI(4)P bound from the VSD-III side show that this is also a stable pose (RMSD <2 Å), where the headgroup interacts with the same positively charged residues as seen for PI(4,5)P₂ (**Fig. 4A** [↗](#), Fig. S6). The residues on the S4-S5 linker, R1463, K1466 and R1469, predominantly bind to the 4'-phosphate (**Fig. 4B** [↗](#)). Due to the more buried location of PI(4)P binding, the PO4 phosphate can associate more with DIII-IV linker lysines (**Fig. 4B** [↗](#)), however, the absence of the 5'-phosphate leads to a reduced number of total electrostatic interactions (Fig. S8).

To investigate structural changes that might occur in the presence of PI(4)P/PI(4,5)P₂, we also carried out simulations of the inactivated Na_v1.4 structure without any PIPs bound for comparison. Although the DIII-IV linker remains bound throughout simulations both with and without PIPs, residues P1336 and R1337 in the DIII-IV linker downstream of the IFM motif are significantly less mobile with PI(4,5)P₂ present (**Fig. 4C** [↗](#)). Additionally, seven residues belonging to the DIV S4-S5 linker, including binding residues K1466 and R1469, also have significantly lower mobility when PI(4,5)P₂ is bound (**Fig. 4C** [↗](#)). In the PI(4)P bound simulations, there were no significant differences in DIII-IV linker or S4-S5 linker mobility.

The PIP binding poses seen in our simulations is similar to resolved binding poses for PI(4,5)P₂ in cryo-EM structures of Ca_v2.2 and K_v7.1 (**Fig. 5A** [↗](#)). A sequence alignment of Na_v1.4 VSDs show that there are more positively charged residues present in the S4/S4-S5 linker regions of DIV compared to the other domains (**Fig. 5B** [↗](#)). PIP is bound at similar VSD residues on both these ion channels, with PIP forming interactions with a gating charge further up the S4 helix in Ca_v2.2 due to the VSD being in a different state. Additionally, the high sequence similarity in the S4/S5-S5 linker and DIII-IV linker regions between the nine human Na_v channel subtypes suggests conservation of the PIP binding site (**Fig. 5C** [↗](#)).

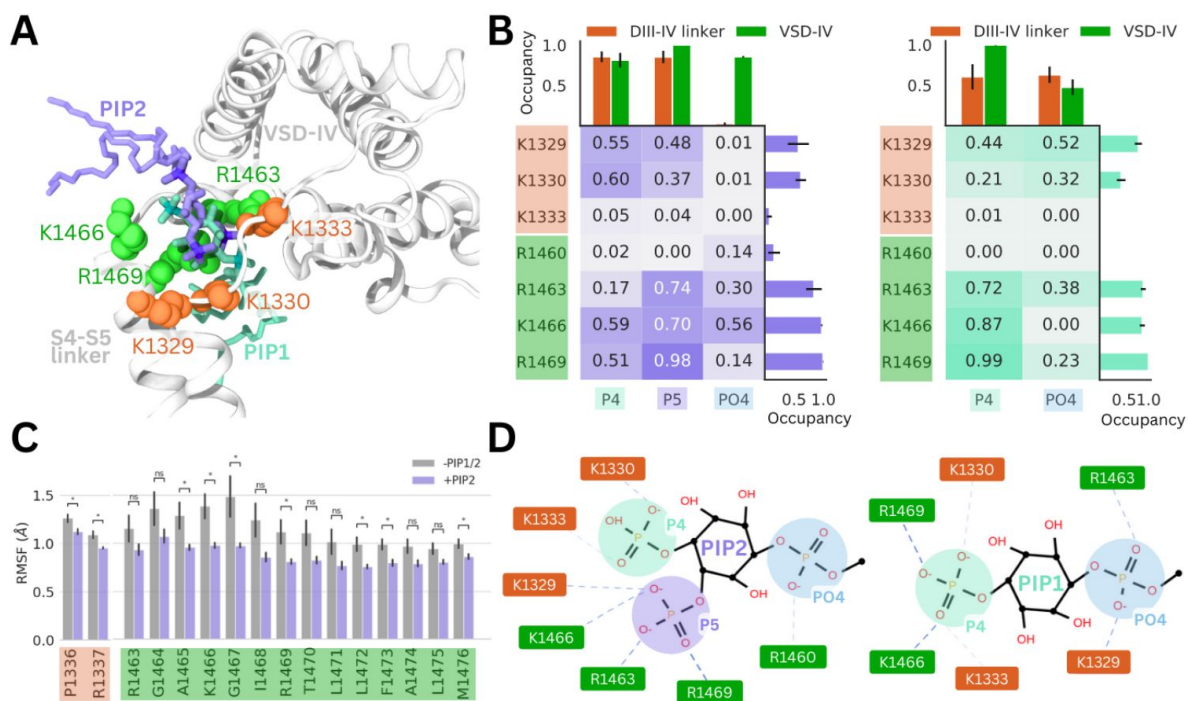


Fig. 4.

PI(4,5)P₂ and PI(4)P binding to Na_v1.4 in atomistic simulations

(A) Representative snapshots of PI(4,5)P₂ bound from the VSD-I side (purple stick) and PI(4)P bound from the VSD-III side (cyan stick), with six basic residues forming the binding site located on the DIII-IV linker (orange VDW representation) and VSD-IV S4-S5 linker (shown in green VDW representation) visualized from the intracellular face of the protein **(B)** Proportion of frames where each of the binding site residues were identified to be within 4.5 Å with the different headgroup regions, P4, P5 and PO4, for PI(4,5)P₂ (left) and PI(4)P (right) **(C)** Comparison of RMSF per carbon-alpha for simulations with and without bound PI(4,5)P₂, showing residues on the S4-S5 linker and DIV linker with significant differences in mobility (p-value < 0.05) **(D)** Interaction network plots between the PIP headgroup and basic binding residues on DIII-IV linker (orange) and DIV S4-S5 linker (green), generated by ProLIF – showing the dominant interactions across simulations of PI(4,5)P₂ and PI(4)P.

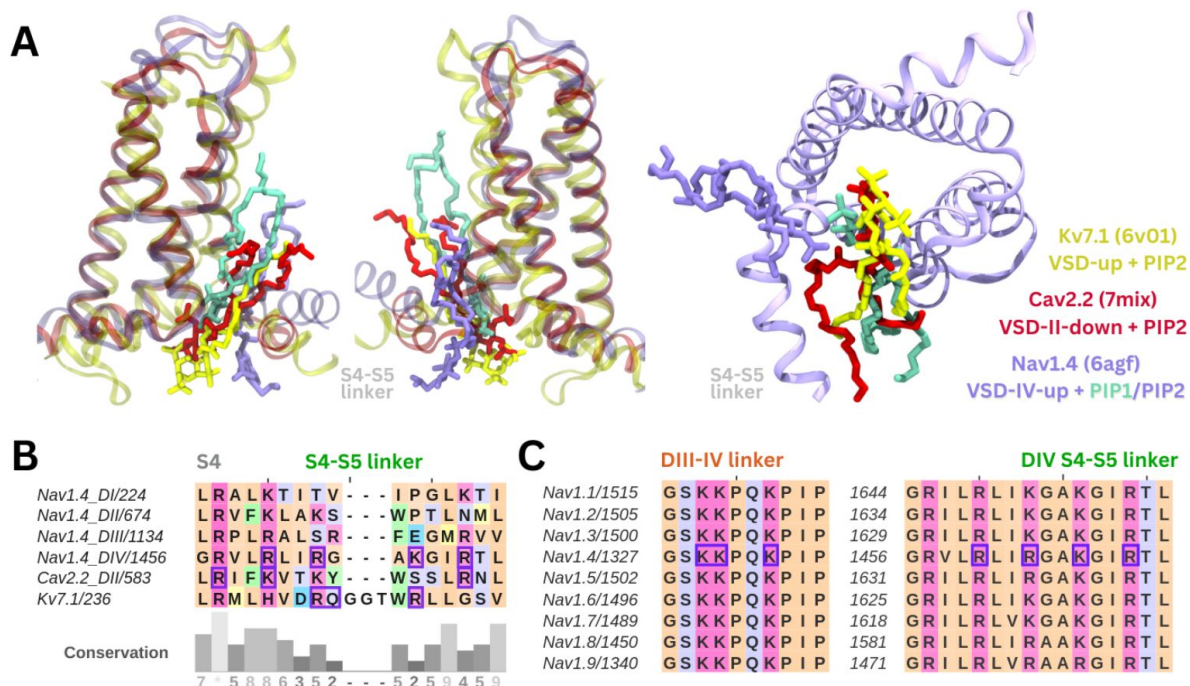


Fig. 5.

Comparison of the identified phosphoinositide binding site to Na_v subtypes and other ion channels

(A) Binding poses of $\text{PI}(4,5)\text{P}_2$ (in purple) and $\text{PI}(4)\text{P}$ (in cyan) aligned with two other tetrameric channels structures $\text{K}_v7.1$ (6v01, in yellow) and $\text{Ca}_v2.2$ (7mix, in red) that were resolved with $\text{PI}(4,5)\text{P}_2$ at their respective VSDs. (B) Sequence alignment of the S4 helix and S4-S5 linker of the four domains of $\text{Na}_v1.4$, compared to VSD-II of $\text{Ca}_v2.2$ and one of the four identical VSDs of $\text{K}_v7.1$; residues colored by amino acid class; purple boxes indicate $\text{PI}(4,5)\text{P}_2$ binding residues (identified with 5 Å of the headgroup). (C) Sequence alignment of the nine human Na_v channel subtypes shows high sequence similarity in the S4 helix, S4-S5 linker and DIII-IV linker regions.

To assess possible state- and subtype-dependent differences in PIP binding, we performed embedded three structures of Nav1.7 with different VSD conformations in a PIP enriched membrane (**Fig. 6A**). The inactivated Nav1.7 structure (blue, PDB ID: 6j8g) contains all VSDs in the activated/up state and a bound DIII-IV linker. We also simulated a NavPas (American cockroach) chimera structure with a human Nav1.7 VSD-IV in the deactivated/down state (VSDs I-III are activated, and from NavPas) (pink, PDB ID: 6nt4). This structure features a dissociated DIII-IV linker and a resolved C-terminal domain bound to the DIV S4-S5 linker (forming the ‘inactivation switch’) at residues identified to form part of our PIP binding site. To assess PIP binding at VSD I-III in the deactivated/down state, we modelled the Na_v1.7 resting state based on templates structures in which different VSDs have been captured in deactivated states with the aid of toxins (detailed in Table S2). These structures feature three or more of the gating charge residues below the hydrophobic constriction site (HCS) and displacement of the S4-S5 linker (**Fig. S9**). Given that no resting state mammalian Na_v channel structure has been resolved, it is possible that the modelled VSDs may not reflect the fully deactivated state.

In triplicate 50 μ s coarse-grained simulations, PIPs bind to the analogous site to that seen in Na_v1.4 in the inactivated Na_v1.7 structure, interacting with residues belonging to both the DIII-IV linker and VSD-IV for durations comparable to Nav1.4, (**Fig. 6B-C**, **Fig. S10**). Binding of PIP to the DIV S4-S5 linker to the deactivated VSD-IV in the Nav1.7-NavPas chimera and resting state model was also observed (**Fig. 6B-C**). However, in the Nav1.7-NavPas chimera, the PIP bound at the S4-S5 linker cannot simultaneously associate with the DIII-IV linker (**Fig. S10**), due to its sequestration by the C-terminal domain, which moves the lysines away from the binding VSD-IV residues (**Fig. 6D**). Instead, K1491, K1492 and K1495 (on the DIII-IV linker) are occupied by different PIPs on the other side of the VSD. In the resting state model, which features an unbound DIII-IV linker PIPs binding at the DIV S4-S5 residues also do not associate with the DIII-IV linker (**Fig S10**). Comparison of binding durations at this site across the three systems reveals a greater number of long PIP interactions ($> 20 \mu$ s) with inactivated Nav1.7 (5) compared to Nav1.7-NavPas (3) or the resting state model (2) where VSD-IV is down, and the DIII-IV linker is dissociated.

There are state-dependent differences in PIP occupancy at the gating charges in each VSD (**Fig. 6E-F**, **Fig. S11**). PIP can associate with the lowest three gating charges VSD-I of the resting state model, but not in the inactivated state (**Fig. 6E**). This is likely due to the large displacement in the DI-S4 helix, which moves down three helical turns in the resting state model so that all gating charges are below the HCS (**Fig. S9**). In VSD II and III, the PIP interaction differences between inactivated and resting state are present but less pronounced, owing to a smaller difference in the relative displacement between the gating charges between states (**Fig. S9**). PIP occupancy is also higher at VSD-IV when it is in the deactivated conformation, however, the presence of a bound CTD in the Nav1.7-NavPas chimera reduces this.

Discussion

Recently, PI(4,5)P₂ was shown to be a negative regulator of Na_v1.4, modulating channel kinetics and voltage dependence. Presence of PI(4,5)P₂ causes a depolarizing shift in the voltage dependence of activation, making Na_v1.4 opening more difficult in a physiological context. Additionally, it stabilizes the inactivated state of Na_v1.4, marked by both shortened inactivation times and slowed recovery from the inactivation.

Using a multiscale simulation approach, we identified a putative PIP binding site comprised of positively charged residues belonging to the S4 helix/S4-S5 linker of VSD-IV (R1460, R1463, K1466, R1469) and DIII-IV linker (K1329, K1330, K1333, K1352). Coarse grained simulations of Na_v1.4 embedded in a complex membrane showed that PIP interacts with residues belonging to VSD-IV and the DIII-IV linker. In coarse grained enriched PIP simulations, PIP2 formed longer duration interactions with Na_v1.4 than PIP1 and PIP3, supported by a greater number of charged

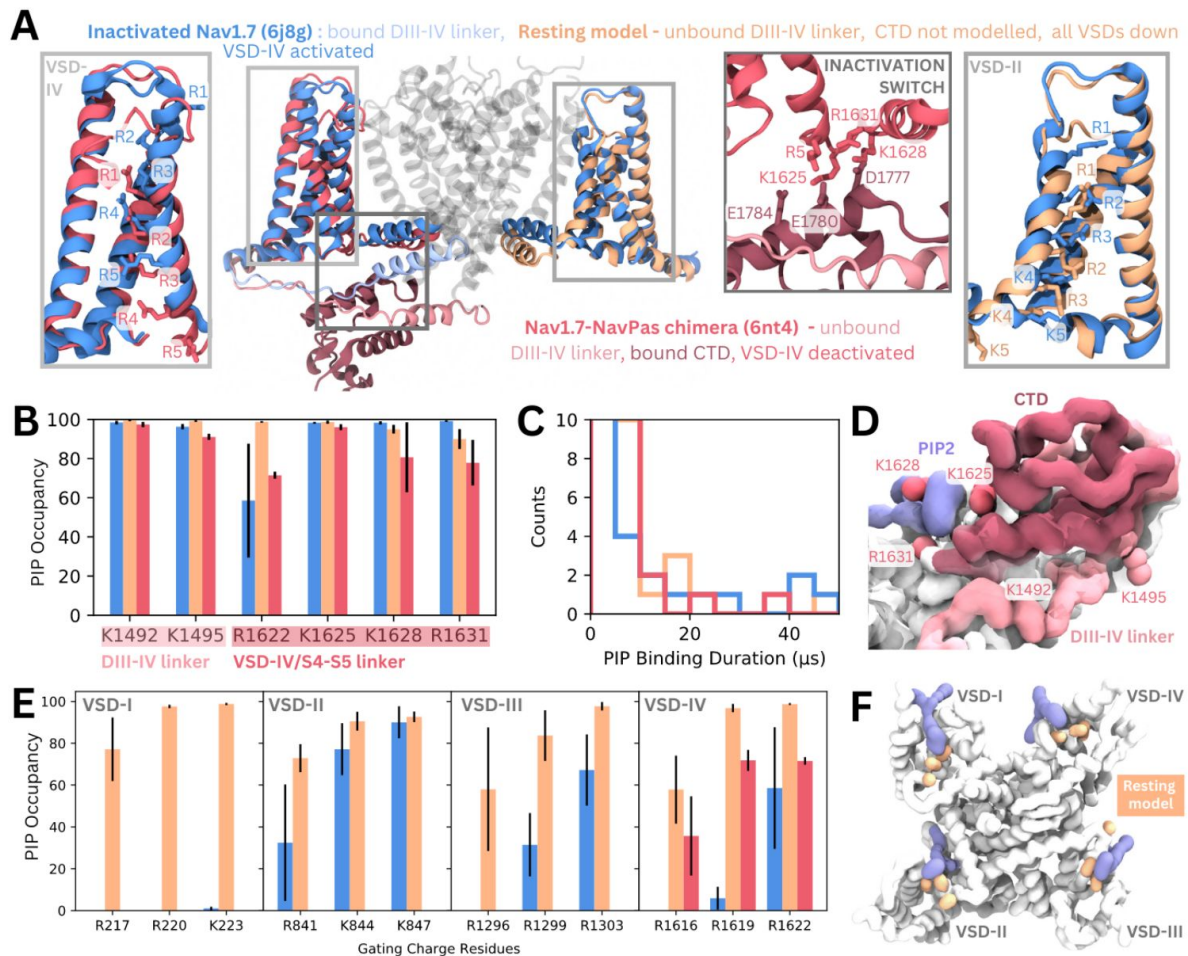


Fig. 6.

PIP binding to Nav1.7 with different VSD states in coarse-grained simulations

(A) Atomistic representation of the three different Nav1.7 structures simulated: (1) the inactivated state (blue, PDB ID: 6j8g) with the VSDs all in the activated, up state (2) the Nav1.7-NavPas chimera (pink, PDB ID: 6nt4) with the Nav1.7 VSD-IV in the deactivated, down state and a bound NavPas CTD (3) a Nav1.7 resting state model (orange, model generation detailed in Table. S2) with all four VSDs in the deactivated, down state. Panel insets show the different conformations of VSD-IV (left) and VSD-II (right) across different structures. The inactivation switch formed by the CTD and VSD-IV S4-S5 linker proposed by Clairfeuille et al. is shown (middle). (B) Occupancy of all PIPs (PIP1, PIP2, PIP3) at binding site residues in the three systems (C) Distribution of PIP binding durations at the identified site (D) Representative simulation snapshot of PIP binding at DIV S4-S5 linker in the Nav1.7-NavPas system. The CTD (dark pink) prevents PIP access to DIII-IV linker lysines (E) PIP occupancy at the bottom three gating charges on VSD I-III in the inactivated (blue) and resting state model (orange) simulations. For VSD-IV, PIP occupancy in the Nav1.7-NavPas system (pink) is also shown. (F) Representative simulation snapshot showing PIP (purple) binding at the gating charges (orange) in the resting state model simulations.

interactions. Atomistic simulations verified the stability of PI(4,5)P₂ (the most common PIP₂ species in the plasma membrane) at this site and showed that the binding of PI(4,5)P₂ reduces the mobility of some DIV S4-S5 and DIII-IV linker residues. Simulations of Na_v1.7 with VSDs in different conformational states showed that the PIP binding site is conserved in Na_v1.7 and that PIP interactions at VSD gating charges are functional state dependent, with more interactions being formed when the VSDs are deactivated.

The DIII-IV linker, CTD and S4-S5 linkers all play key roles throughout the Na_v conformational cycle. Mutation of the IFM inactivation motif as well as other residues in the DIII-IV linker alter fast inactivation and recovery from fast inactivation (McPhee et al., 1998 [↗](#); West et al., 1992 [↗](#)). While the precise role of the Na_v C-terminal domain and its conformation during the Na_v activation cycle remain elusive, it is likely to be important for coordinating fast inactivation (Clairfeuille et al., 2019 [↗](#); Mantegazza et al., 2001 [↗](#)). CTD binding to DIV S4-S5 and sequestration of the DIII-IV linker is proposed to occur in the resting state. After the pore opens, activation of VSD-IV is thought to cause CTD dissociation, releasing the DIII-IV linker to allow fast inactivation. Residues in ‘switch 1’ of the CTD-binding site on the DIV S4-S5 stably bind PIP in our simulations of inactivated Nav1.4 and Nav1.7. We hypothesize that PIP binding at this location makes it more difficult for the CTD to reassociate with VSD-IV, a conformational change which is required during recovery from inactivation.

More generally, the S4-S5 linkers in all four domains couple the VSD to the pore helices and adopt different orientations depending on VSD activation states. When the VSD is activated (in the open and inactivated states), the S4-S5 linkers lie parallel to the membrane. In the resting state, when the voltage sensor is deactivated, the S4-S5 linkers move downward below the plane of the membrane. We propose that PIP binding at the identified site could additionally stabilize both the DIV S4-S5 linker and DIII-IV linker to favor the inactivated state. Although the recovery from fast inactivation occurs on the order of several milliseconds (Bezanilla and Armstrong, 1977 [↗](#)), and is beyond atomistic simulation timescales, we observed statistically significant reductions in the RMSF of several DIII-IV linker and DIV S4-S5 linker residues when PI(4,5)P₂ was bound in 1.5 μs atomistic simulations. Reduction in the mobility of the DIII-IV linker may slow the dissociation of the upstream IFM motif and stabilization of the S4-S5 linker prevents the downward movement required for the channel to transition back to the resting state.

The PIP binding residues identified here are conserved in Na_v1.1-1.9 (**Fig. 5D** [↗](#)), suggestive of a shared binding site and mechanism for PIP-mediated modulation across subtypes. Mutations at these conserved residues in other subtypes lead to various gain-of-function diseases (Table S1). For example, analogous to the Na_v1.4 R1469 residue, the R1642C mutation (in Na_v1.3) leads to developmental epileptic encephalopathy (Zaman et al., 2018 [↗](#)), and R1644C/H mutations (in Na_v1.5) cause cardiac arrhythmias (Frustaci et al., 2005 [↗](#)). These diseases are associated with accelerated rates of channel recovery from inactivation, consistent with our observations that an interaction between PI(4,5)P₂ and the residue corresponding to R1469 in other Na_v subtypes could be important for prolonging the fast-inactivated state.

The PIP binding site identified here harbors sequence and structural similarity to PI(4,5)P₂ binding sites found in other cation channels (**Fig. 5B** [↗](#)). For example, PI(4,5)P₂ is resolved at a similar site near the VSD and S4-S5 linker in a recent cryo-EM structure of K_v7.1, where the phosphate headgroup forms analogous contacts to R249 and R243 (PDB ID: 6v01) (Sun and MacKinnon, 2020 [↗](#)). Despite differences in the role of PI(4,5)P₂, which negatively regulates Na_v1.4 but is required for K_v7.1 pore opening, the binding site appears to be conserved. Based on the PI(4,5)P₂ binding site, a structurally similar compound was developed as an activator of K_v7 channels and proposed to be a future antiarrhythmic therapy (Liu et al., 2020 [↗](#)).

PI(4,5)P₂ also binds to the down, deactivated state of VSD-II in Ca_v2.2 (PDB ID: 7mix) (Gao *et al.*, 2021 [↗](#)). In this structure, the PI(4,5)P₂ headgroup interacts with two VSD-II gating charges, R584 and K587. Compared to the positioning of PI(4,5)P₂ in our simulations of Na_v1.4 with an activated VSD-IV, the headgroup associates further up the S4 helix in Ca_v2.2 due to the VSD being in a deactivated state. This is also seen in K_v7.1 which contains an extended GGT loop in the S4-S5 linker which prevents PI(4,5)P₂ binding in the VSD-down state (Mandala and MacKinnon, 2023 [↗](#)). In our coarse-grained simulations of the resting Na_v1.7 model, we observe a similar state-dependent difference in PI(4,5)P₂ interactions with the deactivated states of each VSD. Since activation of VSDs I-III are known to be coupled to channel opening [\[1\]](#), we propose that PIP binding at these VSDs impedes their ability to activate and thus increases the voltage threshold required of opening.

The leftward shift in voltage dependence of inactivation is less pronounced when PI(4,5)P₂ is converted to PI(4)P rather than completely dephosphorylated to PI (Gada *et al.*, 2023 [↗](#)). This suggests that PI(4)P may play a compensatory role when PI(4,5)P₂ is not present. This is supported by our simulations which show that PI(4)P can also stably occupy the identified binding site, albeit with shorter duration and form less electrostatic interactions compared to PI(4,5)P₂. Our atomistic simulations also showed that the 5'-phosphate is more important than the 4'-phosphate for forming interactions with the DIV S4-S5 linker residues of the inactivation switch. These factors might suggest that PI(4,5)P₂ binding is preferred over PI(4)P at this site and can better compete to bind over the CTD, suggesting that PI(4,5)P₂ is more effective at stabilizing the inactivated state and preventing recovery to the resting state.

Simulations using the Martini2.2 forcefield have previously been used to investigate lipid-protein interactions and successfully predict specific lipid binding sites, including the PIP binding site on Kir channels (Stansfeld *et al.*, 2009 [↗](#)). While Martini2.2 simulations explicitly represent all three PIP species (PIP1, PIP2 and PIP3) present in the mammalian membrane, it does not distinguish between phosphate positions within each charge state (e.g. PI(3,4)P₂ vs PI(4,5)P₂). In this work, we employed atomistic interactions to complement and strengthen findings from coarse grain simulations, allowing us to identify the specific contribution of each PI(4,5)P₂ phosphate group to binding as well as investigate the conformational changes associated with PI(4,5)P₂ binding.

Conclusion

Using multiscale simulations, we show that PI(4,5)P₂ binds stably to inactivated structures of Nav1.4 and Nav1.7 at a conserved site within the DIV S4-S5 linker. As the CTD is proposed to also bind here during recovery from inactivation, we hypothesize that PI(4,5)P₂ competes with the CTD to bind to this site, prolonging inactivation. At this site, PI(4,5)P₂ simultaneously binds to the DIII-IV linker which is responsible for allosterically blocking the pore during fast inactivation (**Fig. 7** [↗](#)). Its binding reduces the mobility of both the DIV S4-S5 and DIII-IV linkers, potentially slowing the conformational changes required for the channel to recover to the resting state. We also propose that in the resting state, PIPs form additional interactions with S4 gating charges, particularly in VSD-1, anchoring them to the membrane in a way which may make the upwards movement required for their activation more difficult. Our results provide insight into how sodium channels are modulated by phosphoinositides, an important step for the development of novel therapies to treat Nav-related diseases.

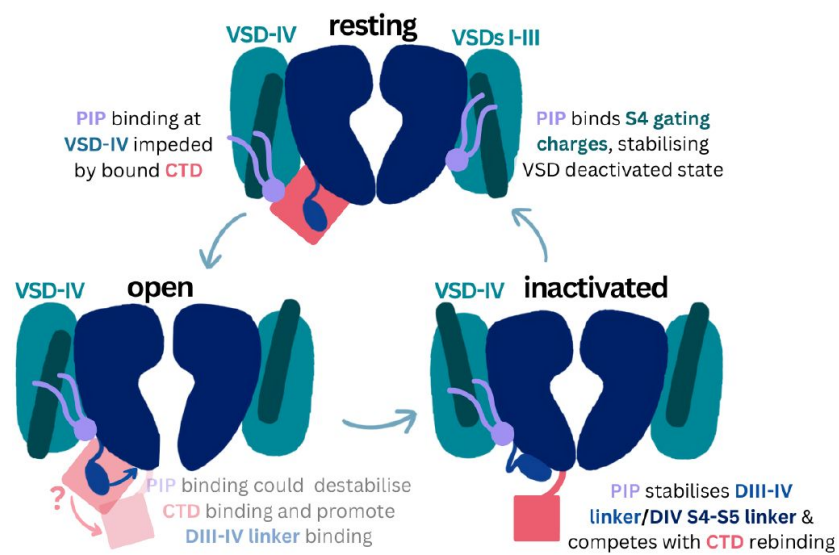


Fig. 7.

Proposed mechanism of PIP effects on the sodium channel functional cycle

Methods

Coarse grained simulations

Coarse-grained simulations of Na_v1.4 embedded in a complex mammalian membrane were carried out to investigate lipid-protein interactions. The inactivated Na_v1.4 alpha subunit (PDB ID: 6agf) (Pan *et al.*, 2018 [DOI](#)) was coarse-grained using the CHARMM-GUI Martini Maker (Jo *et al.*, 2008 [DOI](#); Qi *et al.*, 2015 [DOI](#)) and embedded in a 360 Å x 360 Å complex membrane using insane.py (Wassenaar *et al.*, 2015 [DOI](#)). The composition of the complex mammalian membrane is as reported in Ingólfsson *et al.* (2017) [DOI](#). Three replicate simulations, each with different starting coordinates, were carried out for 16 μs each.

To better sample binding events, we also carried out PIP-enriched simulations in which Na_v1.4 was embedded in a 160 Å x 160 Å POPC membrane with 5% of each PIP species (PIP1, PIP2, PIP3) added to the cytoplasmic leaflet using insane.py. Ten replicate simulations, each with different starting coordinates, were carried out for 80 μs each.

To explore possible state and subtype dependent differences in PIP binding, the inactivated Na_v1.7 structure (PDB ID: 6j8g) (Shen *et al.*, 2019 [DOI](#)) and the Na_v1.7-NavPas chimera with CTD bound and VSD-IV in the deactivated state (PDB ID: 6nt4) (Clairfeuille *et al.*, 2019 [DOI](#)) were also coarse-grained and simulated in PIP-enriched membranes (same protocol as above) for three replicates of 50 μs each. Additionally, a model Na_v1.7 with all four VSDs in the deactivated state using Modeller (Šali and Blundell, 1993 [DOI](#)). The template and structural information for this model are detailed in Table S2. In brief, VSD-I down was modelled from 7xve (Huang *et al.*, 2022 [DOI](#)), VSD-II and VSD-III were both modelled from the deactivated VSD-II from 7k48 (Wisedchaisri *et al.*, 2021 [DOI](#)), and VSD-IV and the unbound DIII-IV linker were modelled from the corresponding regions in 6nt4 (Clairfeuille *et al.*, 2019 [DOI](#)). Adjacent S5/S6 regions to each VSD were also modelled from each specified template to ensure proper contacts between the pore domain and VSDs. The CTD was not included in the model.

All systems were solvated and ionized with 150 mM NaCl. All coarse-grained simulations were carried out with GROMACS 2022 (Bauer *et al.*, 2023 [DOI](#)) using the Martini2.2 forcefield (de Jong *et al.*, 2013 [DOI](#)). Energy minimization was carried out on each system using the steepest descent method for 1,000 steps. Following this, equilibration in the constant pressure, constant volume (NVT) ensemble at 1 atm for 10 ps was carried out with backbone position restraints (1,000 kJ mol⁻¹nm⁻²) using a 2-fs timestep. Following this, constant pressure and temperature (NPT) equilibration simulations were carried out, using 5-, 10-, and 20-fs timesteps in sequence, with each running for 5,000 steps. 1 atm pressure was maintained using a Berendsen barostat with semi-isotropic conditions. Production simulations were carried out in the NPT ensemble, kept at a temperature of 310 K using the Nose-Hoover thermostat (Evans and Holian, 1985 [DOI](#)) and a pressure of 1 bar using the Parrinello-Rahman barostat (Parrinello and Rahman, 1981 [DOI](#)). A time step of 20 fs was used. During production simulations, the backbone beads were weakly restrained to their starting coordinates using a force constant of 10 kJ mol⁻¹nm⁻².

Atomistic simulations

Atomistic simulations were performed to characterize atomic interactions between Na_v1.4 residues and the bound PI(4,5)P₂ headgroup. Frames from a stable PIP2 binding event (from replicate 1 of enriched PIP simulations) were clustered using a selection of the bound PIP2 headgroup beads (C1 C2 C3 PO4 P1 P2) and binding residues K1329, K1330, K1333, K1463, K1466 and R1469 with an RMSD cutoff of 2.5 Å. The protein and bound PIP2 were extracted from the representative frame of the cluster and backbone beads of the coarse-grained VSD-IV were aligned to the corresponding carbon-alpha atoms in the original cryo-EM structure of Na_v1.4 (Pan *et al.*, 2018 [DOI](#)). PIP2 was backmapped to atomistic coordinates of SAPI24 (the CHARMM lipid for PI(4,5)P₂ with protonation at the 4-position phosphate) and the protein was replaced with the 6agf

structure. The system was embedded in a 140 Å x 140 Å POPC membrane, solvated and 0.15 M NaCl added using the CHARMM-GUI Membrane Builder (Jo *et al.*, 2008 [↗](#); Lee *et al.*, 2016 [↗](#); Wu *et al.*, 2014 [↗](#)). An identical system was set up with Na_v1.4 in a POPC membrane without PIP.

Atomistic simulations were performed with Amber20 (Case *et al.*, 2020 [↗](#)), using the CHARMM36m (Huang *et al.*, 2017 [↗](#)) and TIP3P water (Jorgensen *et al.*, 1983 [↗](#)) forcefields. Equilibration steps were performed (minimisation, heating, pressurising), with 5 kJ/mol restraints on the protein backbone, followed by 24 ns of gradually reducing restraints. Five replicates of unrestrained production equilibrium simulations were performed, run for 1.5 µs each. The temperature was set at 310 K using the Langevin thermostat (Loncharich *et al.*, 1992 [↗](#)) and at a collision frequency of 5 ps⁻¹. Pressure was set at 1 bar using the Monte Carlo barostat (Åqvist *et al.*, 2004 [↗](#)) with anisotropic scaling and relaxation time of 1 ps. 12 Å van der Waals cut-off and hydrogen bond SHAKE constraints were used. Hydrogen mass repartitioning was used to enable a 4 fs timestep (Hopkins *et al.*, 2015 [↗](#)).

Analysis

Simulations were visualised and protein image figures produced using Visual Molecular Dynamics (VMD) (Humphrey *et al.*, 1996 [↗](#)).

Coarse grain lipid-protein interactions were characterized using in-house python scripts which used the numpy, MDAnalysis (Michaud-Agrawal *et al.*, 2011 [↗](#)) and pandas libraries (as done previously in Lin *et al.* (2022) [↗](#)). A cut-off of 0.7 nm was used to define interactions between lipids and protein residues. Distance heatmaps were generated based on the minimum distance between a PIP bead and the side chain (SC2) bead of the arginine or lysine residue interacting. Binding durations were calculated by first counting the number of interactions between each PIP and VSD-IV binding residues (R1460, R1463, K1466 and R1469). A binding event is defined as the time between the first and last time that PIP interacts with two or more binding residues on VSD-IV, provided that it interacts with a minimum of one binding residue between this period.

For atomistic simulations, MDAnalysis (Michaud-Agrawal *et al.*, 2011 [↗](#)) was used to calculate RMSD of various parts of the protein and PIP headgroup, and RMSF of the carbon-alphas. Statistical significance in RMSF was assessed using student's t-test. ProLIF (Bouysset and Fiorucci, 2021 [↗](#)) was used to compute electrostatic interactions between the protein and binding residues. Representative snapshot of PIP2 binding was generated using the WMC Clustering Tool in VMD to identify the top cluster of the PIP2 headgroup (RMSD cutoff of 3 Å), then subsequently cluster the six binding residues to identify the most representative binding configuration. Trajectories were strided every 1 ns and the first 250 ns of simulations was discarded as equilibration time for analyses of RMSF, ProLIF interactions and clustering.

ClustalOmega (Goujon *et al.*, 2010 [↗](#); Sievers *et al.*, 2011 [↗](#)) and JalView (Waterhouse *et al.*, 2009 [↗](#)) were used to generate and visualize sequence alignments. Structural representations of K_v7.1, Ca_v2.2 and Na_v1.4 structures were created in VMD by aligning each of the S4 helices on the VSD where the PIP was bound.

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Reviewer #1 (Public Review):

Summary: Here, the authors were attempting to use molecular simulation or probe the nature of how lipids, especially PIP lipids, bind to a medically-important ion channel. In particular, they look at how this binding impact the function of the channel.

Strengths: the study is very well written and composed. The techniques are used appropriately, with plenty of sampling and analysis. The findings are compelling and provide clear insights into the biology of the system.

Weaknesses: a few of the analyses are hard to understand/follow, and rely on "in house" scripts. This is particularly the case for the lipid binding events, which can be difficult to compute accurately. Additionally, a lack of experimental validation, or coupling to existing experimental data, limits the study.

It is my view that the authors have achieved their aims, and their findings are compelling and believable. Their findings should have impacts on how researchers understand the functioning of the Nav1.4 channel, as well as on the study of other ion channels and how they interact with membrane lipids.

- <https://doi.org/10.7554/eLife.91218.1.sa2>

Reviewer #2 (Public Review):

Summary:

Y., Tao E., et al. used multiscale MD simulations to show that PI(4,5)P₂ binds stably to an inactivated state of Nav channels at a conserved site within the DIV S4-S5 linker, which couples the voltage sensing domain (VSD) to the pore. The authors hypothesized that PI(4,5)P₂ prolongs inactivation by binding to the same site where the C-terminal tail is proposed to bind during recovery from inactivation. They convincingly showed that PI(4,5)P₂ reduces the mobility of both the DIV S4-S5 linker and the DIII-IV linker, thus slowing the conformational changes required for the channel to recover to the resting state. They also conducted MD simulations to show that phosphoinositides bind to VSD gating charges in the resting state of Nav channels. These interactions may anchor VSD at the resting state and impede its activation. Their results provide a mechanism by which phosphoinositides alter the voltage dependence of activation and the recovery rate from inactivation, an important step for developing novel therapies to treat Nav-related diseases. However, the study is incomplete and lacks the expected confirmatory studies which are relevant to such proposals.

Strengths:

The authors identified a novel binding between phosphoinositides and the VSD of Nav and showed that the strength of this interaction is state-dependent. Based on their work, the affinity of PIPs to the inactivated state is higher than the resting state. This work will help pave the way for designing novel therapeutics that may help relieve pain or treat diseases like arrhythmia, which may result from a leftward shift of the channel's activation.

Weaknesses:

However, the study lacks the expected confirmatory studies which are relevant to such proposals. For example, one would expect that the authors would mutate the positive residues that they claim to make interactions with phosphoinositides to show that there are much fewer interactions once they make these mutations. Another point is that the authors found that the main interaction site of PIPs with Nav1.4 is the VSD-DIV and DIII-DIV linker, an interaction that is expected to delay fast inactivation if it happens at the resting state. The authors should make a resting state model of the Nav1.4 channel to explain the recent experimental data showing that PIP₂ delays the activation of Nav1.4, with almost no effect on the voltage dependence of fast inactivation.

Major concern:

1- Lack of confirmatory experiments, e.g., mutating the positive residues that show a high affinity towards PIPs to a neutral and negative residue and assessing the effect of mutagenesis on binding.

2- Nav1.4 is the only channel that has been studied in terms of the effect of PIPs on it, therefore the authors should build a resting state model of Nav1.4 and study the effect of PIPs on it.

Minor points:

There are a lot of incorrect statements in many areas, e.g., "These diseases 335 are associated with accelerated rates of channel recovery from inactivation, consistent with our observations that an interaction between PI(4,5)P2 and the residue corresponding to R1469 in other Nav 337 subtypes could be important for prolonging the fast-inactivated state." Prolonging the fast inactivated state would actually reduce recovery from inactivation and not accelerate it.

- <https://doi.org/10.7554/eLife.91218.1.sa1>

Reviewer #3 (Public Review):

Summary:

This work uses multiscale molecular dynamics simulations to demonstrate molecular mechanism(s) for phosphatidylinositol regulation of voltage gated sodium channel (Nav1.4) gating. Recent experimental work by Gada et al. JGP 2023 showed altered Nav1.4 gating when Nav1.4 current was recorded with simultaneous application of PI(4,5)P2 dephosphorylate. Here the authors revealed probable molecular mechanism that can explain PI(4,5)P2 modulation of Nav1.4 gating. They found PIP lipids interacting with the gating charges - potentially making it harder to move the voltage sensor domain and altering the channels voltage sensitivity. They also found a stable PIP binding site that reaches the D_IV S4-S5 linker, reducing the mobility of the linker and potentially competing with the C-terminal domain.

Strengths:

Using multiscale simulations with course-grained simulations to capture lipid-protein interactions and the overall protein lipid fingerprint and then all-atom simulations to verify atomistic details for specific lipid-protein interactions is extremely appropriate for the question at hand. Overall, the types of simulation and their length are suitable for the questions the authors pose and a thorough set of analysis was done which illustrates the observed PIP-protein interactions.

Weaknesses:

Although the set of current simulations and analysis supports the conclusions drawn nicely, there are some limitations imposed by the authors on the course-grained simulations. If those were not imposed, it would have allowed for an even richer set and more thorough exploration of the protein-lipid interactions. The Martini 2 force field indeed cannot change secondary structure but if run with a properly tuned elastic network instead of backbone restraints, the change in protein configuration can be sampled and/or some adaptation of the protein to the specific protein environment can be observed. Additionally, with the 4to1 heavy atoms to a bead mapping some detailed chemical specificity is averaged out but parameters for different PIP family members do exist - including specific PIP(4,5)P2 vs PIP(3,4)P2, and could have been explored.

- <https://doi.org/10.7554/eLife.91218.1.sa0>

Author Response

We thank the reviewers for their helpful comments and thorough assessment of our manuscript which will allow us to improve the work in a subsequent revision. Many

suggestions, such as mutating residues to help validate the proposed site will be included in a future revision. Below we clarify three aspects that led to confusion in the initial review

The comment of reviewer 2 that "... the main interaction site of PIPs with Nav1.4 is the VSD-DIV and DIII-DIV linker, an interaction that is expected to delay fast inactivation if it happens at the resting state." is true. However, as explained in our manuscript (Fig. 7), we don't expect binding at this position to happen in the resting state as the C-terminal domain is bound to this region, impeding PIP binding.

Reviewer 2 also suggests that we produce a resting state model of Nav1.4 to replace/supplement the results we obtained using our resting Nav1.7 model. We chose to model Nav1.7 due to the availability of structures with different VSDs in the deactivated conformation, something that is not true for Nav1.4. While we plan to explore a Nav1.4 resting state based on the reviewer's suggestion, we note that this introduces an extra layer of uncertainty. However, due to sequence conservation of the gating charges and proposed binding site residues between Nav subtypes, we propose very similar modes of PIP binding among the Nav subtypes across the different conformations.

Finally, we strongly disagree with the reviewer's assessment that there are 'There are a lot of incorrect statements in many areas' and this may have come from a misreading of the mentioned sentence. The sentence in question reads "These diseases 335 are associated with accelerated rates of channel recovery from inactivation, consistent with our observations that an interaction between PI(4,5)P₂ and the residue corresponding to R1469 in other Nav 337 subtypes could be important for prolonging the fast-inactivated state." To which the reviewer 2 states 'Prolonging the fast inactivated state would actually reduce recovery from inactivation and not accelerate it.' The statement quoted is not incorrect – from the original experiments we know that the presence of PIP prolongs the time spent in the fast inactivated state. Mutations at the PIP binding site are likely to reduce PIP binding, and with less PIP present the channel will recover from inactivation more quickly. We appreciate that this sentence could be reworded for clarity and will address this in our revision to prevent such misreading.